

EXPERIMENT 2

Data Modelling and Form Design in Zoho Creator

1. Aim

To explore data modelling and form creation in Zoho Creator by organizing application data, building interactive forms, applying input validation, and managing records with a low-code platform.

2. Introduction

2.1 Data Modelling

Data modelling involves arranging application information into entities, attributes, and relationships. A well-planned model helps maintain consistency, reduce duplicate information, and support efficient data handling. In Zoho Creator, forms, reports, field types, and lookup fields can be used to represent these relationships.

2.2 Form Design in Zoho Creator

Form design focuses on creating a convenient interface for entering and managing information. Zoho Creator provides visual controls and a drag-and-drop builder through which fields, validation conditions, and other form properties can be configured with limited programming.

3. Objectives of the Study

- To understand the basic principles of application data modelling.
- To identify and use suitable field types in Zoho Creator.
- To construct forms using visual design components.
- To apply validation rules for reliable data entry.
- To connect related information using lookup fields.
- To study how forms can support automated business processes.

4. Procedure

- Log in to the Zoho Creator platform.
- Start a new application for the required use case.
- Identify the entities, attributes, and relationships needed for the application.
- Create the required form with the visual form builder.
- Add fields such as Text, Number, Date, Email, Phone, Choice, and Lookup.
- Set field properties and configure suitable validation conditions.
- Save and preview the form to check its appearance and behavior.
- Enter sample records and verify that the data is stored correctly.
- Make necessary changes to the form and validation settings.
- Publish the application after completing the verification.

5. Detailed Analysis

Zoho Creator provides a structured method for handling application data. Different field types help capture information in an appropriate format, while lookup fields can connect related forms. Validation rules help reduce incomplete or invalid entries. Since much of the interface is configured visually, routine application development can be completed with less hand-written code.

6. Illustrative Example

Consider a Student Registration System in which student details are collected through a Zoho Creator form.

6.1 Implementation using Zoho Creator

- Create a Student Registration form.
- Add fields such as Student Name, Register Number, Department, Email, Phone Number, Date of Birth, and Gender.
- Mark important fields as mandatory and apply suitable validation.
- Store submitted student records in the application database.
- Generate reports to view and manage student information.

7. Advantages and Limitations

7.1 Advantages

- Visual and simple form-building process.
- Reduced development time for common applications.
- Less manual programming is required.
- Validation features can be configured within forms.
- Forms and applications can be updated easily.

7.2 Limitations

- Complex customization may require Deluge scripting.
- Applications depend on the capabilities of the selected platform.
- Large or highly specialized enterprise systems may need additional development.

8. Result

The experiment was completed successfully by studying data modelling and creating an interactive form in Zoho Creator. The exercise demonstrated how structured fields, relationships, validation, and visual form design can be combined to collect and manage application data with reduced coding effort.

Illustrative Form Design

The image shows a web application interface for a 'student information system'. The main content area is titled 'academic record' and contains a form with the following fields:

- Student name:** A text input field with a placeholder '1'.
- Phone no:** A text input field with a placeholder '+91 81234 56789'.
- Address:** Two text input fields labeled 'Address Line 1' and 'Address Line 2'.
- City / District:** A text input field.
- State / Province:** A dropdown menu with a 'Select' button.
- Postal Code:** A text input field.
- Country:** A dropdown menu with a 'Select' button.
- Enrollment date:** A text input field with a placeholder 'YYYYMMDD'.
- GPA:** A text input field with a placeholder '00.0000-1.0000'.
- Drop Down:** A dropdown menu with a 'Select' button.
- Multi Line:** A text area for multiple lines of text.

The form is displayed within a browser window. The browser's address bar shows the URL 'localhost:3000/academic-record'. The browser's title bar shows 'student information system'. The browser's status bar shows 'That expires in 1 day'.